

ABHA, Saudi Arabia

WMO: 411120

Lat: 18.240N

Lon: 42.657E

Elev: 2090

StdP: 78.61

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	7.9	-12.9	1.6	15.7	-9.9	2.1	18.2	10.4	16.6	9.6	16.8	0.8	330	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	12.4	32.0	13.7	31.1	13.4	30.2	13.5	20.1	24.5	19.5	23.9	18.9	23.4	5.3	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	17.8	22.3	18.1	16.9	22.0	17.2	15.9	21.6	68.7	24.5	66.3	23.9	63.9	23.4	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.3	7.4	3.6	33.5	1.4	1.1	2.6	34.2	1.8	34.9	0.9	35.5	-0.1	36.3		
(5)			-1.2	21.4	1.5	1.1	-2.3	22.2	-3.1	22.8	-4.0	23.4	-5.1	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	14.1	15.9	17.6	19.6	22.2	24.4	24.4	24.0	22.8	19.4	16.4	14.5		
	DBStd	4.02	1.62	1.66	1.83	1.75	1.76	1.33	1.58	1.40	1.41	1.45	1.46	1.35		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	425	130	70	35	8	1	0	0	0	0	6	58	117		
	CDD10.0	3516	129	165	236	288	379	431	446	433	384	291	193	141		
	CDD18.3	898	0	1	12	46	122	181	188	174	134	39	2	0		
	CDH23.3	7035	5	18	84	292	928	1574	1443	1252	1103	304	27	3		
(13)	CDH26.7	2086	0	0	3	24	239	576	494	425	306	19	0	0		
(14)	Wind	WSAvg	3.1	3.5	3.8	4.1	3.4	2.7	2.9	3.0	3.0	2.6	2.5	2.8		
(15) Precipitation	PrecAvg	191	11	9	28	43	29	8	18	25	7	3	6	2		
	PrecMax	501	50	99	344	224	98	66	76	121	39	24	37	18		
	PrecMin	43	0	0	0	1	1	0	1	4	0	0	0	0		
	PrecStd	107	14	21	65	55	26	13	15	24	9	7	10	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	25.1	26.9	28.7	31.8	33.1	33.2	32.2	31.9	28.2	25.2	23.8		
		MCWB	10.7	12.0	13.1	13.7	13.7	13.9	14.2	14.4	13.5	12.0	11.9	9.7		
	2%	DB	22.2	23.9	25.2	27.2	30.2	31.9	31.9	31.2	30.6	27.1	24.0	22.1		
		MCWB	11.4	12.4	12.9	13.5	13.3	13.2	13.8	13.9	13.2	11.4	12.1	11.2		
	5%	DB	20.9	22.8	24.1	26.1	29.1	31.0	30.8	30.2	29.6	26.1	23.1	21.1		
		MCWB	11.5	12.5	13.1	13.9	13.7	12.9	13.8	14.0	12.9	11.2	12.1	11.5		
(25) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	19.3	21.2	23.0	25.1	28.0	30.0	29.5	29.2	28.5	25.1	22.0	20.0		
		MCWB	11.7	12.4	13.4	14.2	14.1	13.1	14.2	14.5	12.6	11.2	12.2	11.9		
	0.4%	WB	15.4	16.2	17.6	18.2	19.9	20.3	20.7	21.0	20.0	17.9	17.1	15.7		
		MCDB	19.5	20.5	21.6	22.4	24.3	25.1	25.3	25.2	23.9	20.7	20.4	19.3		
	2%	WB	14.5	15.5	16.5	17.4	18.9	19.4	19.7	20.2	18.9	17.0	16.2	14.9		
		MCDB	18.5	19.8	20.9	22.2	23.5	24.5	24.5	24.4	22.8	20.3	19.5	18.6		
(31) Mean Daily Temperature Range	5%	WB	13.8	14.7	15.8	16.7	18.1	18.6	19.0	19.6	18.0	16.2	15.5	14.3		
		MCDB	17.5	19.1	20.3	21.9	22.9	24.0	23.7	23.8	22.0	19.6	18.9	17.8		
	10%	WB	13.3	14.2	15.2	16.1	17.2	17.5	18.2	18.9	17.0	15.4	14.9	13.7		
		MCDB	16.8	18.4	19.8	21.2	22.9	23.6	23.1	23.1	22.0	19.2	18.3	17.1		
	5% DB	MDBR	10.7	10.9	10.4	11.0	12.0	12.4	11.5	12.0	12.6	12.5	11.8	11.7		
		MCDBR	13.8	13.1	12.4	12.3	13.0	13.1	12.4	12.8	13.0	13.1	13.1	13.6		
(36) Clear-Sky Solar Irradiance	taub	taud	0.330	0.367	0.400	0.441	0.500	0.556	0.615	0.602	0.499	0.392	0.370	0.335		
		Ebn at Noon	955	932	909	869	810	757	715	730	807	893	901	936		
	Edh at Noon	taud	109	133	153	173	204	237	257	253	206	146	127	110		
		RadAvg	5.06	5.81	6.57	6.89	7.19	7.23	6.58	6.34	6.56	6.34	5.36	4.96		
(45)	All-Sky Solar Radiation	RadStd	0.27	0.30	0.31	0.38	0.35	0.24	0.28	0.21	0.19	0.22	0.19	0.28		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.63	+0.49	N/A	+0.91	N/A	N/A	-105	+277	+173
(47)	Regional (4 neighbors)	+0.47	N/A	N/A	+0.49	N/A	N/A	-11	+192	+154

Nomenclature: See separate page